Benha University
Faculty of Engineering (Shoubra)
Electrical Engineering

ELE123 Technical Reports

ELE114 Technical Report Writing

(2026-2027)

Dr. Christina Gamal

Course grade

Code	Subject	Contact Hours				Marks				Duration of Final Examination (hours)
		Lec.	Tut.	Lab.	Total	Sem. Work	Lab./Oral Exam	Written Exam	Total	
ELE123	Technical Reports	2	0	0	2	15	15	30	60	2

Code	Subject	Contact Hours				Marks				Duration of Final Examination
		Lec.	Tut.	Lab.	Total	Sem. Work	Lab./Oral Exam	Written Exam	Total	
ELE114	Technical Report Writing	1	0	2	3	30	15	45	90	2

Course Content

- Introduction to Technical Report Writing
- Basic Report Structure
- Preparing Technical Reports using MS Word
- Using Styles in MS Word for Technical Reports
- Graphical Representation of Data
- Cover Letter and CV
- Business Letter, Memorandum and Email
- Latex

Introduction to Technical Report Writing

Lecture 1

Lecture Outlines

1. What is the Technical Writing Report?
2. Purpose of Technical Writing Report
3. Readers for Technical Writing Report
4. Types of Documents in Technical writing
5. Characteristic of Technical Reports
6. Plagiarism Definition
7. How to Maintain Ethical Standards in Technical Writing
8. Guidelines for Good Writing
9. Technical writing Process

1- What is the Technical Writing Report?

- In engineering, one of the major forms of communication is the technical report.
- **Technical writing** conveys specific information about a technical subject to a specific reader for a specific purpose.
- It communicates technical information, facts, and conclusions.
- It is divided into sections that allow different readers to access different levels of information.
- The ability to produce a clear, concise, and professionally presented report is a skill you need to develop.

2- Purpose of Technical Writing Report

➢ Every report has a purpose. Some common purposes are:

- To **convince** the reader of something. For example:
  - ❖ to convince the public that a proposed project will bring benefits
- To **encourage** the reader to do something. For example:
  - ❖ to encourage an organization to co-operate with your company on a project
- To **inform** the reader about something. For example:
  - ❖ to present the outcomes of a project to stakeholders

3- Readers for Technical Writing Report

- Three broad categories of Readers for a technical report can usually be identified from the beginning:

1. Technical

are made up of people in the same field as yours. They are people such as other engineers, researchers, technical specialists, or scientists. They may be people you know within your own organization, or they may work outside. They have strong knowledge of the field, being familiar with technical terminology, equations, standards, and detailed analyses, and focusing on methodology, calculations, data, and results.

2. Non-Technical

may include for example, citizen, financial experts, legal or administrative readers, and others. Their needs will be quite different from your "technical" readers. As they read your report, they are looking for certain things. They have limited or no technical knowledge of the subject. They require simple explanations and minimal use of complex technical terminology. They are interested in the overall objective, benefits, and outcomes.

3. Mixed

are the most common readers including both technical and non technical readers. Such readers are more difficult to write for.

3- Readers for Technical Writing Report … Cont.

- There is one thing nearly all your readers have in common.

They are usually busy and have a great deal of reading to do every day. If you hand them a thick, boring, complicated, and badly written report you will bother them immediately.

Overcoming this natural reaction is a challenge, but you can make it easier for them by design your report properly.

Once the subject of your report has been identified and its purpose defined, your first task is to analyze your reader.

4- Types of Documents in Technical writing

Major types of documents in technical writing can be grouped into four major categories:

i. Educational / Academic Writing

- Papers, theses, books, magazine articles, and scripts.

ii. Reports for day-to-day business

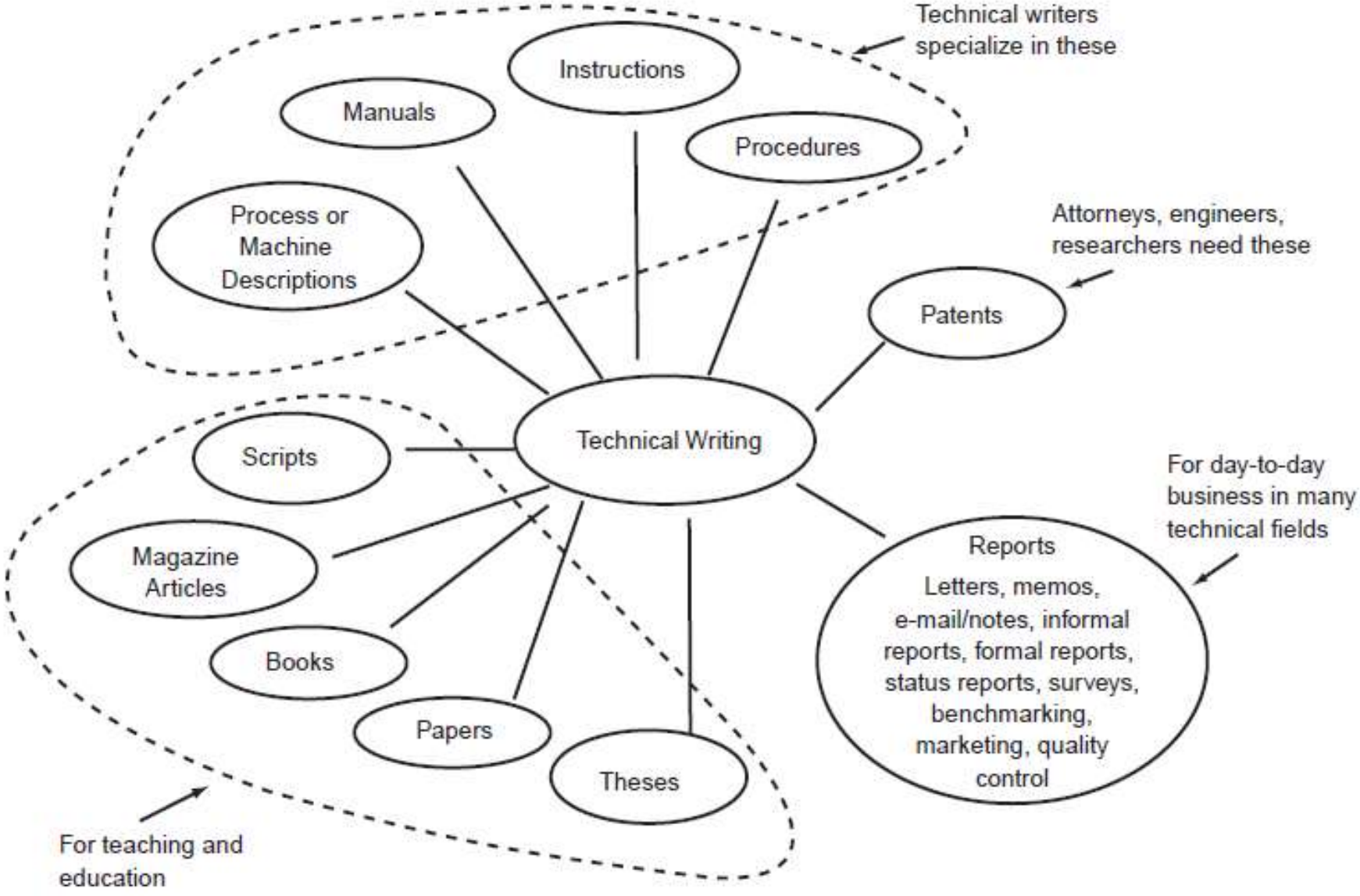
- Includes letters, memos, emails/notes, status reports, surveys.

iii. Patents

- Legal documents that describe new inventions.

iv. Instructions, Procedures, Manuals

- A manual showing how to operate a machine.



Spectrum of technical writing

Spectrum of Technical Writing

5- Characteristic of Technical Reports

The four C's:

- ***Clear*** *-- it is easily understood by your intended reader*
- ***Comprehensive*** *-- all of the necessary information is present*
- ***Conciseness*** *-- it is clear without excess verbiage*
- ***Correct*** *-- it is grammatical and follows conventions*

6- Plagiarism Definition

Plagiarism can be defined as follows:
To use another person's words or ideas as if they were your own.

The following are seen as plagiarism:

- To steal another person's work
- To pay another person to write your assignment
- To copy directly from a source without referencing the original source
- To use another person's ideas without giving credit to the original ideas
- To copy another person's work word for word

7- How to Maintain Ethical Standards in Technical Writing

Make sure you have the information you need to write accurate reports by avoiding the following items:

✓ Avoid plagiarism by properly citing all sources and giving credit to original authors.

- **Incorrect (Plagiarism):** "antenna gain is related to both directivity and efficiency."

☞ (Copied from a textbook or article without citation.)

 Correct: According to Balanis (2016), antenna gain is related to both directivity and efficiency.

✓ Use accurate and verified information from reliable sources.

- **Incorrect:** The operating frequency of Wi-Fi is always 2.4 GHz."

☞ (This is inaccurate because Wi-Fi also operates at 5 GHz and 6 GHz.)

 Correct: Wi-Fi commonly operates in the 2.4 GHz, 5 GHz, and 6 GHz frequency bands according to IEEE 802.11 standards.

✓ Do not suppress important information.

- **Incorrect:** "The proposed algorithm achieved 95% accuracy.

☞(The writer does not mention that the algorithm was tested on only 50 samples.)

 Correct: "The proposed algorithm achieved 95% accuracy when tested on a dataset of 50 samples; further testing on larger datasets is needed to validate its performance.

7- How to Maintain Ethical Standards in Technical Writing … Cont.

✓ Do not emphasize incorrect information.

- **Incorrect way:** "Don't connect the red wire to the blue terminal."

☞ The reader may remember *red wire* and *blue terminal* but forget the "don't."

- **Better way:** "Always connect the red wire to the red terminal."

☞ The instruction is positive, clear, and emphasizes the **correct action**.

✓ Avoid language that could possibly mislead readers.

- **Incorrect:**: "The new system will completely eliminate network problems."
- **Better way:** "The new system is expected to reduce network problems by approximately 30%."

✓ Avoid hiding part of information that mislead others to follow up your research.

- **Misleading:** "Press the button to start."
- **Clear:** "Press the green power button on the right side of the device to start it."

8- Guidelines for Good Writing

- *Write positive language.*
  - Negative: "Do not forget to submit the form."
  - Positive: "Remember to submit the form."

- *Don't write in upper case for more than A WORD OR TWO.*
  - Bad: "PLEASE READ ALL INSTRUCTIONS CAREFULLY BEFORE STARTING."
  - Good: "Please read all instructions carefully before starting."
- *Use lists whenever possible.*
  - Bad: "To log in, enter your username, then your password, and finally click login."
  - Good:

    To log in:

    1.Enter your username.
    2.Enter your password.
    3.Click Login
- *Use headings and subheadings.*
- *Use the title to define the task.*
  - Bad title: "Guide."
  - Good title: "How to Install the Printer Driver."

8- Guidelines for Good Writing … Cont.

➢ *Be precise with your instructions.*

- Bad: "Heat the liquid for a while."
- Good: "Heat the liquid at 70°C for 15 minutes."

➢ *Use warnings.*

Use warnings when there is a possibility of damage, danger, or an important mistake.
⚠ **Warning:** Always unplug the device before cleaning to prevent electric shock.

➢ *Write a conclusion.*

A conclusion should briefly summarize the main findings or results of the report.

- Bad: Report ends suddenly after results.
- Good: "In conclusion, the new method reduces costs by 20% and improves efficiency."

➢ *Don't annoy the reader.*

Avoid unnecessary repetition, complicated sentences, and irrelevant information.

- Bad: "The results are very important. These important results show that the system has important improvements."
- Good: "The results show that the system improved performance."

➢ *Keep your paragraphs short (Less is better).*

- Bad: A long paragraph of 12–15 sentences containing several different ideas without breaks.
- Good: A short paragraph of about 3–5 sentences that focuses on one main idea.

9- Technical writing Process



End of Lecture

Best Wishes